

Visual Question Answering From Charts

Nagesh Kumar, Adarsh Ramesh Thimmapurmath

School of Computing, Dublin City University

Dublin, Ireland

{nagesh.kumar2, adarshramesh.thimmapurmath2}@mail.dcu.ie

Abstract—The literature on visual question answering (VQA) for charts focuses on datasets, methods, and challenges in the field. Key datasets such as PlotQA, ChartQA, DVQA, and Chart-to-Text address issues related to synthetic data, fixed-vocabulary responses, and complex reasoning tasks. These datasets emphasize the need for handling numerical operations, multimodal interactions between visual, textual, and structured data, and managing out-of-vocabulary (OOV) answers. Various approaches, including hybrid OCR models, transformer-based architectures (T5, VisionTaPas), and multimodal frameworks (ChartAssistant, DePLOT, MATCHA), have been explored to improve chart comprehension by incorporating chart-to-table translation, mathematical reasoning, and large language models (LLMs).

Traditional VQA models struggle with real-world chart interpretation, highlighting the importance of specialized training on chart-related tasks and structured data extraction. Notably, MATCHA achieves a 20% improvement on ChartQA through targeted methodologies. However, challenges such as color-based reasoning, high-precision calculations, and complex visual references persist. To address these, research emphasizes multi-modal fusion, explainability, and robust data de-rendering. Future directions include enhancing numerical reasoning, expanding datasets to cover diverse chart types, and bridging the gap between simulated and real-world applications. These advancements are essential for improving model architectures, evaluation standards, and practical applications in data analysis, decision-making, and explainable AI.

I. INTRODUCTION

Visual question answering proves to be very important in connecting computer vision and natural language processing. It enables robots to interpret visual information and answers question from people. Conventional VQA modeled challenges have been on the understanding of natural images only. However, more recent developments have gone into specialized areas, for instance, understanding charts and plots, as this is a key skill in any other intelligence for an organization, data analysis automation, and the development of explainable AI systems. Charts and plots encode very complex numerical, relational, and contextual information, and present difficult problems that are other than the typical VQA. These include out of vocabulary responses, arithmetic and logical operations, and transformation of visual properties, such as color and position, to a structured format.

The more robust benchmarks PlotQA, ChartQA, and Chart-to-Text—came into being to respond to the various shortcomings of the early datasets, namely FigureQA and DVQA, that relied on artificial charting, given vocabulary, and simple questions. With millions of question-answer pairs in

these datasets collected from authoritative sources like the World Bank and Statista, the datasets actually exemplified the complexity of the real world, thus calling for multi-step reasoning over numerical and visual elements. However, the cutting-edge advanced models still struggle with reasoning based on colors, certain highly precise numerical calculations, and generalizations across different chart types: Transformer-based architecture (T5, VisionTaPas), hybrid pipeline (OCR + classification), and multimodal framework (ChartAssistant, DePLOT).

The literature review aims to explore three key research questions related to the development of chart-focused Visual Question Answering (VQA). First, it examines how existing datasets can address the limitations of synthetic data and fixed-vocabulary responses in chart comprehension. Second, it investigates which architectural advancements, such as pretraining for mathematical reasoning or translating charts into tables, can improve model performance on complex visual-language tasks. Finally, it identifies the persistent challenges that hinder practical applications and explores potential strategies to bridge these gaps in the future.

This work has synthesized advances in dataset designs (PlotQA’s 28.9M OOV-oriented QA pairs), model architecture (like MATCHA’s gain of 20% precision due to derendering charts) and evaluation metrics (for example, DePLOT’s relative mapping similarity) from the initial articles. There are also areas of concern, such as that important computational cost of pretraining multimodal systems and loss of visual context under table-based reasoning. Although the results are intended to guide researchers toward improvements in numeric reasoning, wide-ranging methods of multimodal fusion, and matching synthetic benchmarks to the variation in real-world visualizations. This review helps to ensure that ongoing innovation in the design and evaluation frameworks of models stems from the ability of chart-oriented VQA to access scalable, interpretable data analysis tools.

II. RELATED RESEARCH

A. Dataset

Previous studies have emphasized advancing chart comprehension through datasets (shown in Tabel 1) and models that address real-world complexity, such as handling out-of-vocabulary (OOV) answers, numerical reasoning, and multimodal integration. [Methani et al., 2020], [Masry et al., 2022],

and [Kafle et al., 2018] prioritize large-scale, diverse question-answer pairs to challenge models with tasks like arithmetic operations and chart-specific terminology. These datasets share a focus on bridging synthetic and real-world scenarios, though their implementations differ: PlotQA [Methani et al., 2020] leverages real-world data sources (e.g., World Bank) for numerical OOV challenges, while ChartQA [Masry et al., 2022] balances human-written analytical questions with machine-generated templated queries to ensure breadth and depth.

[Methani et al., 2020] presents an extensive collection of 28 million question-answer pairs extracted from 224K scientific plots. This dataset is designed to challenge models with numerical reasoning and out-of-vocabulary answers. In a similar way, [Masry et al., 2022] compiles 20,882 charts accompanied by over 32,000 questions—crafted both by humans and generated automatically—to support tasks requiring arithmetic and logical reasoning. Focusing on bar charts, [Kafle et al., 2018] assembles more than 3 million image-question pairs, which emphasize the need to process chart-specific text. Adding further diversity, [Kim et al., 2020] utilizes data from 52 real-world charts along with hundreds of human-generated questions and visual explanations, thereby enriching the dataset with interpretability aspects. Meanwhile, [Meng et al., 2024] leverages the ChartSFT dataset to cover multiple chart types, supporting tasks such as summarization and numerical reasoning. Complementary to these, [Liu et al., 2022a] and [Liu et al., 2022b] build upon datasets like PlotQA and ChartQA to convert visual plots into structured tables and incorporate mathematical reasoning into the learning process. For retrieval and summarization tasks, [Nowak et al., 2024] adapts existing chart datasets, and [Kantharaj et al., 2022] offers a benchmark consisting of 44,096 charts for generating natural language summaries. Finally, although primarily text-based, [Rajpurkar, 2016] has significantly influenced multimodal research by providing over 100,000 question-answer pairs that set a high standard for precise span extraction.

Dataset	Size	Types
PlotQA	224,377 plots	Line, bar, scatter plots
ChartQA	20,882 charts	Bar, line, pie
DVQA	3M image	Bar charts only
Chart-to-Text	44,096 charts	Bar, line, pie, radar, box plots

Table1: Comparison between existing datasets

B. Data Extraction

Previous studies by PlotQA [Methani et al., 2020] cannot be considered as conclusive because it leverages a TableQA framework to tackle open-vocabulary questions, yet it neglects the integration of visual chart elements—such as color and layout which are important for visual reasoning. As the authors note earlier, more work is necessary to unify data tables and visual attributes for tasks like arithmetic operations. This gap is addressed by [Masry et al., 2022] which explicitly incorporates visual references into benchmark, enabling models to reason

over both data tables and visual attributes for tasks like arithmetic operations. Many models rely on structured data extraction (e.g., OCR, table parsing) to translate visual charts into machine-readable formats. Hybrid models [Methani et al., 2020] and [Liu et al., 2022a] exemplify this approach, combining OCR with table extraction for QA tasks. However, innovations diverge in their emphasis: [Liu et al., 2022b] integrates math reasoning pretraining to enhance numerical accuracy, whereas [Meng et al., 2024] adopts a multitask framework for zero-shot generalization across chart types. Transformer-based architectures like T5 and VisionTaPas [Masry et al., 2022] unify visual and textual features, while [Kafle et al., 2018] DVQA employs dynamic encoding to handle chart-specific vocabularies.

A recurring challenge across studies is the loss of visual context (e.g., color, spatial relationships) during data extraction, which impacts performance on tasks requiring color-based reasoning or positional analysis. While [Liu et al., 2022a] DePLOT achieves high accuracy in table reconstruction (97.1% RNSS), it struggles with color-dependent queries, a limitation also noted in MATCHA [Liu et al., 2022b]. Similarly, Chart-to-Text [Kantharaj et al., 2022] and Text-to-Text [Raffel et al., 2020] models generate fluent summaries but face hallucinations and factual inaccuracies, reflecting broader issues in balancing fluency and precision.

In contrast, SQuAD [Rajpurkar, 2016], though text-focused, underscores the universal gap between human and machine comprehension, with machines achieving 51% F1 versus humans at 86.8%. This mirrors challenges in chart VQA [Kafle et al., 2018], where even state-of-the-art models like DePLOT + Codex (67.6% accuracy on ChartQA) fall short of human-like robustness.

The papers collectively advocate for multimodal fusion and specialized pretraining as pathways forward. Yet, their strategies vary: some prioritize architectural innovations (e.g., MATCHA’s [Liu et al., 2022b] derendering), while others focus on expanding dataset diversity (e.g., Chart-to-Text’s 44,096 charts). These efforts highlight a shared vision of interpretable, scalable chart comprehension tools, tempered by the need to address persistent gaps in visual context preservation and high-precision reasoning. Papers that survey VQA more broadly [Ishmam et al., 2024] also highlight multiple datasets centered on tasks ranging from synthetic imagery to interactive 3D environments, illustrating diverse yet complementary pathways for advancing multimodal question answering research

C. Model

The modeling approaches in these studies are equally varied. In [Methani et al., 2020], a hybrid model integrates a classification pipeline with a multi-stage process—combining visual element detection, OCR, and structured table extraction—to address both fixed and open-vocabulary queries. [Masry et al., 2022] employs transformer-based architectures VisionTaPas

which is extension of TaPas for QA over charts that blend visual features with extracted numerical data to perform arithmetic and logical operations. In [Kafle et al., 2018], a dual-network design enhanced by dynamic encoding is introduced to effectively manage the unique text elements within bar charts. [Kim et al., 2020] proposes a pipeline that transforms visual queries into data-based ones and generates template-driven visual explanations, thus enhancing interpretability. [Meng et al., 2024] implements a two-stage training method that first converts charts into tables and then applies multi-task instruction tuning, ensuring improved performance across diverse tasks. Further advancing these methods, [Liu et al., 2022a] utilizes an end-to-end Transformer to convert plots into linearized tables, empowering one-shot reasoning by large language models, while [Liu et al., 2022b] enriches visual language pretraining by incorporating explicit mathematical reasoning alongside chart derendering. Additional studies such as [Nowak et al., 2024] and [Kantharaj et al., 2022] integrate OCR-based extraction, vision-language models, and neural techniques inspired by image captioning to support effective chart retrieval and summarization. The foundational approaches described in [Rajpurkar, 2016] continue to influence these models by demonstrating the efficacy of precise text span extraction.

Underlying these innovations are challenges that persist across the various tasks. [Methani et al., 2020] highlights the difficulty in performing intricate numerical reasoning and detecting complex visual elements, particularly when responses extend beyond a fixed vocabulary. The approaches in [Masry et al., 2022] and [Kafle et al., 2018] reveal that even with advanced transformer and dual-network models, operations that require nested computations and the extraction of nuanced, chart-specific text remain demanding. The system described in [Kim et al., 2020], while enhancing interpretability with visual explanations, still encounters issues with simpler query types and natural language fluency. [Meng et al., 2024] demonstrates that aligning graphical elements with implicit numerical data is a continuous hurdle, and while [Liu et al., 2022a] achieves high accuracy in table reconstruction, some critical visual details—such as color—can be lost in translation. In [Liu et al., 2022b], the integration of mathematical reasoning yields notable performance improvements, yet high-precision computations and the preservation of layout nuances continue to present challenges. Furthermore, the retrieval methods in [Nowak et al., 2024] and the summarization strategies in [Kantharaj et al., 2022] illustrate that incorporating multiple modalities can enhance performance, even as these approaches must navigate the complexity of diverse chart formats. Finally, the text comprehension framework established in [Rajpurkar, 2016] underscores the ongoing gap between machine and human understanding in the realm of precise information extraction.

Together, these works weave a comprehensive narrative that advances our understanding of how machines can interpret, analyze, and explain complex visual data. Each study—whether by introducing novel datasets, pioneering innovative models,

or identifying critical challenges—adds a unique dimension to the evolving field of visual and language reasoning.

D. Challenges

Current research reveals several key challenges that persist in the domain of visual question answering and chart reasoning. In [Methani et al., 2020], models are challenged by the need to perform complex numerical reasoning while accurately detecting intricate visual elements, especially when confronted with out-of-vocabulary responses. Similarly, [Masry et al., 2022] identifies difficulties in handling nested operations and integrating arithmetic with logical reasoning, which remain problematic even for advanced transformer-based architectures. [Kafle et al., 2018] further underscores the issue by demonstrating that processing chart-specific text in bar charts requires more robust and dynamic language modeling techniques. Moreover, the pipeline proposed in [Kim et al., 2020]—which focuses on generating visual explanations—still struggles with simpler query types and achieving natural language fluency in its output.

Enhancements are required across multiple dimensions to advance the state-of-the-art. [Meng et al., 2024] suggests that there is considerable room for improvement in aligning graphical elements with implicit numerical data, indicating a need for models that can seamlessly integrate multi-modal inputs without losing critical contextual information. Although [Liu et al., 2022a] demonstrates impressive accuracy in converting plots to structured tables (with table reconstruction accuracies reaching 97.1% and 94.2% on specific metrics), the loss of vital visual attributes such as color and layout calls for enhanced methods that preserve these details. [Liu et al., 2022b] takes an important step by incorporating mathematical reasoning into visual pretraining; however, high-precision computations and effective preservation of layout nuances remain challenging. In the realms of retrieval and summarization, as explored in [Nowak et al., 2024] and [Kantharaj et al., 2022], integrating multiple modalities to produce factually correct and coherent natural language summaries is still an open problem. Furthermore, the text-based comprehension techniques illustrated by [Rajpurkar, 2016] indicate that even in comparatively simpler domains, a significant gap exists between machine and human performance.

Looking ahead, further enhancements are essential. Developing more integrated models that can concurrently handle visual, textual, and numerical data without compromising on any front is a key direction. Techniques that preserve crucial visual attributes—such as color, spatial layout, and contextual metadata—would greatly improve the fidelity of chart-to-table and chart-to-text translations. Additionally, incorporating advanced reasoning modules capable of nested, high-precision computations, perhaps through innovative pre-training paradigms or hybrid architectures, could help bridge the performance gap between current models and human-level understanding. These improvements are crucial for pushing the boundaries of what current visual reasoning systems can

achieve, ultimately enabling more accurate and interpretable machine understanding of complex visual data.

III. CONCLUSIONS

Advancements in visual question answering and chart reasoning rely on comprehensive datasets that capture the complexity of real-world visualizations and on innovative models that integrate visual, textual, and numerical information to drive sophisticated reasoning. A more systematic and theoretical analysis is required for understanding the underlying mechanisms that fuse these diverse modalities. As the authors note earlier, more work is necessary to refine the extraction techniques that convert intricate visual elements into structured data. Additional studies to understand more completely the key tenets of multimodal data fusion are required, while the unexpected findings signal the need for additional studies to understand more about preserving critical visual attributes—such as color and spatial layout—during data transformation. This work addresses the comprehensive integration of diverse data types and reasoning approaches, so far lacking in the scientific literature. A new approach is therefore needed for developing models that balance complexity with interpretability and high-precision reasoning, and one of the tough challenges for all researchers in this domain is achieving seamless alignment between robust datasets and innovative model architectures without compromising the fidelity of visual information.

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